MATH3881/5881: Statistical Machine Learning Theory

Week 5: Theory Tutorial Solutions: Four Optimisers on a Constant Slope

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Worked solutions to the Week 5 Theory Tutorial. Numbering matches the problem sheet, and notation follows its Background section: $\Delta\theta_i^{(t)} := \theta_i^{(t+1)} - \theta_i^{(t)}$ is the per-coordinate step, $g_i^{(t)} := \partial C/\partial\theta_i$ is the gradient component on coordinate i, and the constant-slope regime assumes $g_i^{(t)} = g_i$ (fixed) for all t past some point, giving $v_i^{(t+1)} \approx g_i$ and $s_i^{(t+1)} \approx g_i^2$ in steady state.

Problem 1: Two equivalences on a constant slope

Part (a): Momentum step $\approx$ GD step

Step 1. From Week 4 (§2.1), the Momentum update expresses the step as $\Delta\theta_i^{(t)} = -\alpha\, v_i^{(t+1)}$, where $v_i^{(t+1)} = \beta\, v_i^{(t)} + (1-\beta)\, g_i^{(t)}$.

Step 2. On a constant slope, the Background gives $v_i^{(t+1)} \approx g_i$ for large t.

Step 3. Substituting,

$$\Delta\theta_i^{(t)} \;=\; -\alpha\, v_i^{(t+1)} \;\approx\; -\alpha\, g_i,$$

which is precisely the GD step $-\alpha\, g_i^{(t)}$ evaluated at the same constant gradient.

Part (b): Adam step $\approx$ RMSProp step

Step 1. From Week 4 (§3.2), the RMSProp step is

$$\Delta\theta_i^{(t)} \;=\; -\frac{\alpha\, g_i^{(t)}}{\sqrt{s_i^{(t+1)}} + \varepsilon}.$$

On a constant slope the Background gives $s_i^{(t+1)} \approx g_i^2$, hence $\sqrt{s_i^{(t+1)}} \approx |g_i|$, so

$$\Delta\theta_i^{(t)} \;\approx\; -\frac{\alpha\, g_i}{|g_i| + \varepsilon} \;\approx\; -\alpha\,\frac{g_i}{|g_i|} \;=\; -\alpha\,\mathrm{sign}(g_i),$$

using $|g_i| \gg \varepsilon$ to drop ε from the denominator in the second approximation. Call this the RMSProp steady-state step.

Step 2. From Week 4 (§3.3), the Adam step uses the bias-corrected running averages

$$\hat{v}_i^{(t+1)} \;:=\; \frac{v_i^{(t+1)}}{1-\beta^{t+1}}, \qquad \hat{s}_i^{(t+1)} \;:=\; \frac{s_i^{(t+1)}}{1-\gamma^{t+1}},$$

and is given by

$$\Delta\theta_i^{(t)} \;=\; -\frac{\alpha\,\hat{v}_i^{(t+1)}}{\sqrt{\hat{s}_i^{(t+1)}}+\varepsilon}.$$

Step 3. For large t, the steady-state approximations $v_i^{(t+1)} \approx g_i$, $s_i^{(t+1)} \approx g_i^2$ apply, and the correction factors $1-\beta^{t+1}, 1-\gamma^{t+1} \to 1$. Hence

$$\hat{v}_i^{(t+1)} \;\approx\; g_i, \qquad \hat{s}_i^{(t+1)} \;\approx\; g_i^2, \qquad \sqrt{\hat{s}_i^{(t+1)}} \;\approx\; |g_i|.$$

Step 4. Substituting into the Adam step,

$$\Delta\theta_i^{(t)} \;\approx\; -\frac{\alpha\,g_i}{|g_i|+\varepsilon} \;\approx\; -\alpha\,\mathrm{sign}(g_i),$$

identical to the RMSProp steady-state step derived in Step 1.

Problem 2: What kind of quantity is α?

Part (a): Chain rule

Step 1. The definition $\tilde{\theta}_i := \kappa\,\theta_i$ inverts to $\theta_i = \tilde{\theta}_i/\kappa$, so $\partial\theta_i/\partial\tilde{\theta}_i = 1/\kappa$.

Step 2. Since $\tilde{C}(\tilde{\theta}) = C(\theta)$, the chain rule gives

$$\tilde{g}_i \;=\; \frac{\partial\tilde{C}}{\partial\tilde{\theta}_i} \;=\; \frac{\partial C}{\partial\theta_i}\cdot\frac{\partial\theta_i}{\partial\tilde{\theta}_i} \;=\; g_i\cdot\frac{1}{\kappa} \;=\; \frac{g_i}{\kappa}.$$

Part (b): Comparing the steps under the rescaling

Step 1. From Problem 1, the steady-state per-coordinate step in the original coordinates is

$$\text{GD:}\quad \Delta\theta_i^{(t)} = -\alpha\,g_i, \qquad \text{RMSProp:}\quad \Delta\theta_i^{(t)} \approx -\alpha\,\mathrm{sign}(g_i).$$

Step 2 (GD). Apply the GD rule in the new coordinates, using $\tilde{g}_i = g_i/\kappa$ from part (a):

$$\Delta\tilde{\theta}_i^{(t)} \;=\; -\alpha\,\tilde{g}_i \;=\; -\alpha\,\frac{g_i}{\kappa} \;=\; \frac{1}{\kappa}\,\Delta\theta_i^{(t)}.$$

GD's step in the new coordinates is $1/\kappa$ times the original.

Step 3 (RMSProp). Apply the RMSProp steady-state rule in the new coordinates:

$$\Delta\tilde{\theta}_i^{(t)} \;\approx\; -\alpha\,\mathrm{sign}(\tilde{g}_i) \;=\; -\alpha\,\mathrm{sign}\!\left(\frac{g_i}{\kappa}\right) \;=\; -\alpha\,\mathrm{sign}(g_i) \;=\; \Delta\theta_i^{(t)},$$

using $\kappa > 0$ to drop the positive factor from the sign. RMSProp's step in the new coordinates is unchanged.

Part (c): One sentence

Because RMSProp's per-coordinate step is invariant under coordinate rescaling, a single α produces the same step magnitude regardless of the natural scale of each parameter, so Adam's default $\alpha = 10^{-3}$ works out of the box across architectures whose parameters live at different scales; GD's step scales inversely with the coordinate scale, so a good α depends on the parameter scale of the problem and must be re-tuned every time.